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IE 2004 Computer Networks

03 – Physical Layer

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Lesson outline

- Physical layer functions and components
- Digital signals – Encoding
- Analog signals – Modulation
- Transmission media
  - Copper: UTP
  - Copper: Coaxial
  - Fiber
  - Wireless (free space)
- Transmission impairments
  - Attenuation
  - Interference
  - Crosstalk

What is the Physical Layer

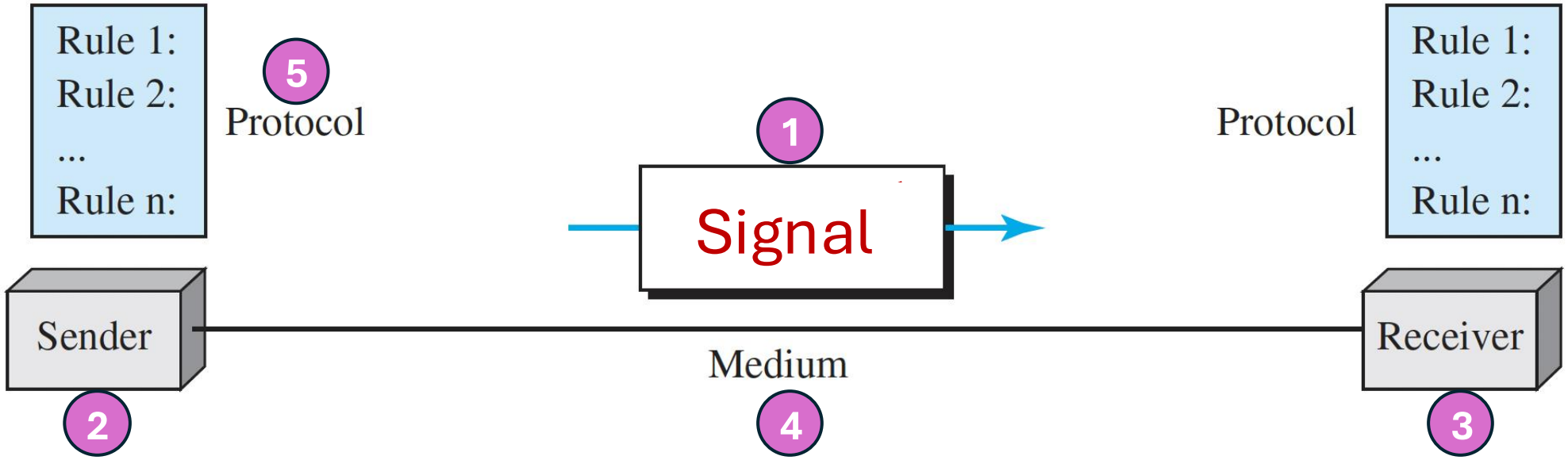
- Lowest layer of the TCP/IP model
- Handles physical transmission of data (1s and 0s)
- Deals with hardware elements and physical connections
- Encompasses cables, connectors, network interface cards, etc.
- Responsible for transmitting signals that represent data over the communication medium
- Physical Layer Standards address two functional areas:
  - Physical Components
  - Encoding / Modulation – Representing 1s and 0s with an electrical or an electromagnetic signal

Physical Components

- Before any network communications can occur, a physical connection to a local network must be established.
- This connection could be wired or wireless, depending on the setup of the network
- A Network Interface Card (NIC) connects a device to the network
- Some devices may have just one NIC, while others may have multiple NICs (Wired and/or Wireless, for example)
- Not all physical connections offer the same level of performance
- Hardware components like NICs, interfaces and connectors, cable materials, and cable designs are all specified in the physical layer standards

Communication model

- Data: Message represented using 1s and 0s
- Signals: Electrical or electromagnetic representations of data
- It is the signal (representation of the data) that travels over the transmission medium, not the message or data



Definition of signals in communication

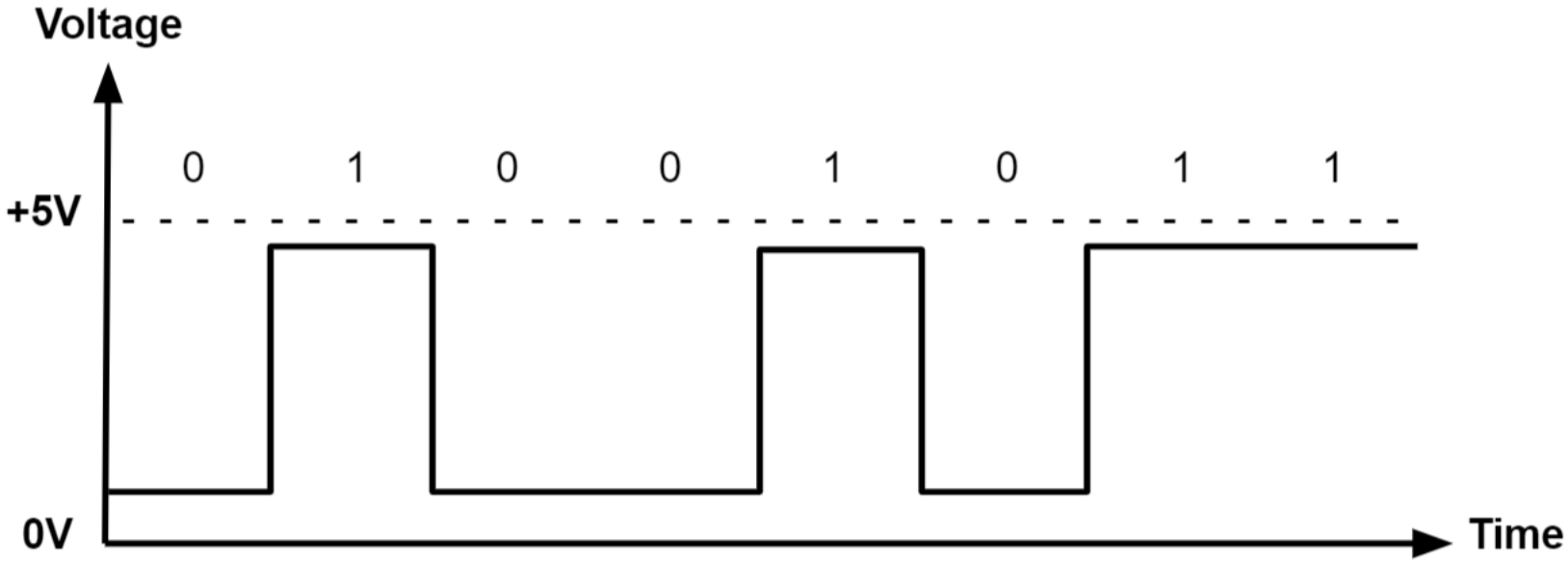
- Signals are representations of data (1s and 0s)
- Signals carry information from one point to another over the transmission medium
- Signals can be in different forms:
  - Electrical (voltage, current) – over copper cabling
  - Electromagnetic: Includes Optical (light) – over free space, optical fiber
- A signal can be analog or digital
  - Encoding: Representing data with a digital signal
  - Modulation: Representing data with an analog signal
  - Decoding: Extracting data from the received digital signal
  - De-modulation: Extracting data from the received analog signal

Bit period and Bit rate

- Bit period is the time taken to transmit a single bit
- If the Bit rate (transmission rate) is R bits per second (bps)
  Bit period = 1/R
- Examples
  - If R = 10 Mbps, bit period = 1 / (10 x 10^6) = 0.1 µs (microseconds)
  - If R = 100Mbps, bit period = 1 / (100 x 10^6) = 0.01 µs
  - If R = 1Gbps, bit period = 1 / (1 x 10^9) = 0.001 µs = 1 ns (nanoseconds)

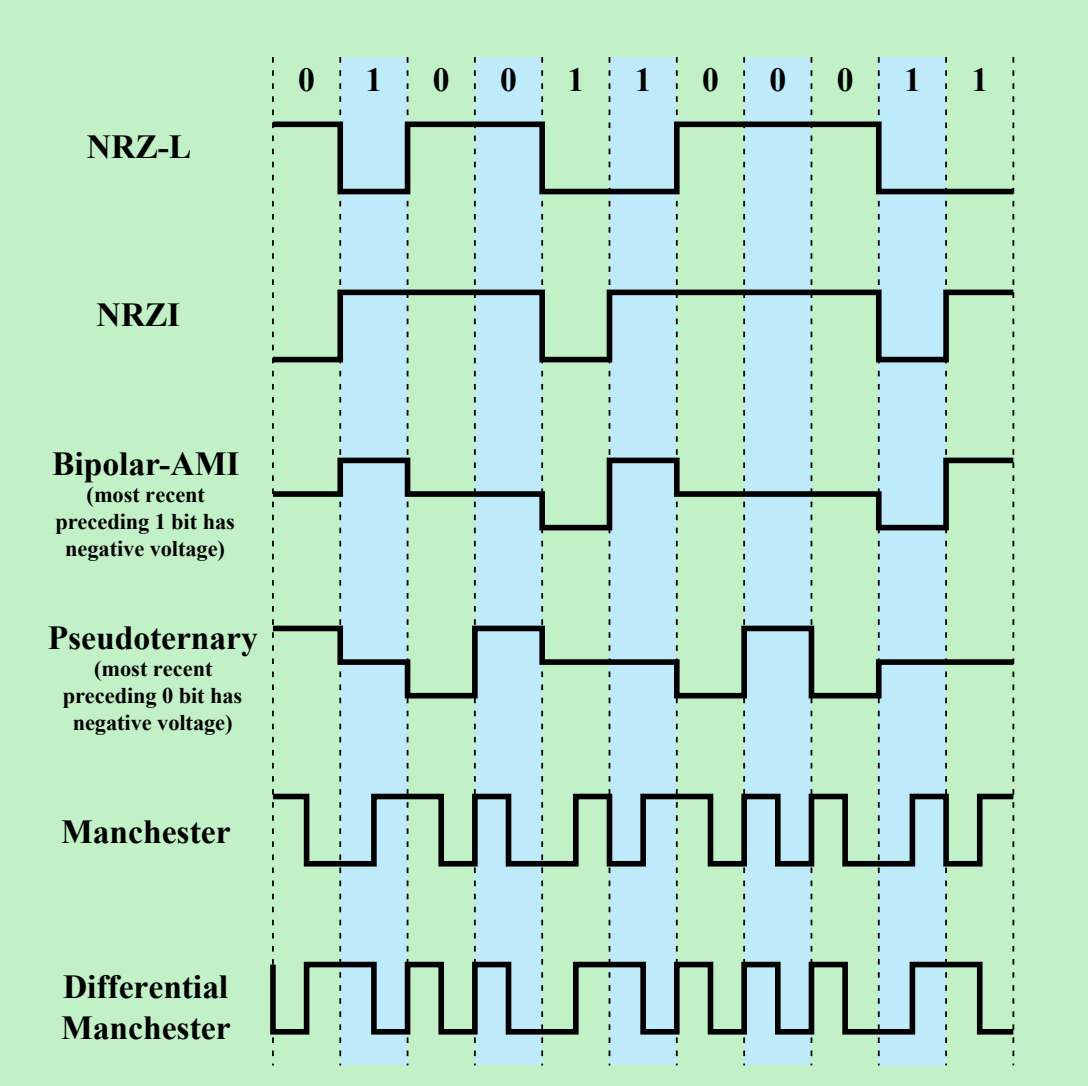
Digital Signals

- Digital signals represent data using discrete values or symbols
- Sequence of discrete voltage pulses
- Each pulse/voltage level is a signal element
- Binary data are transmitted by encoding each data bit into signal elements
- Example:
  binary 0 –> 0V
  binary 1 –> +5V



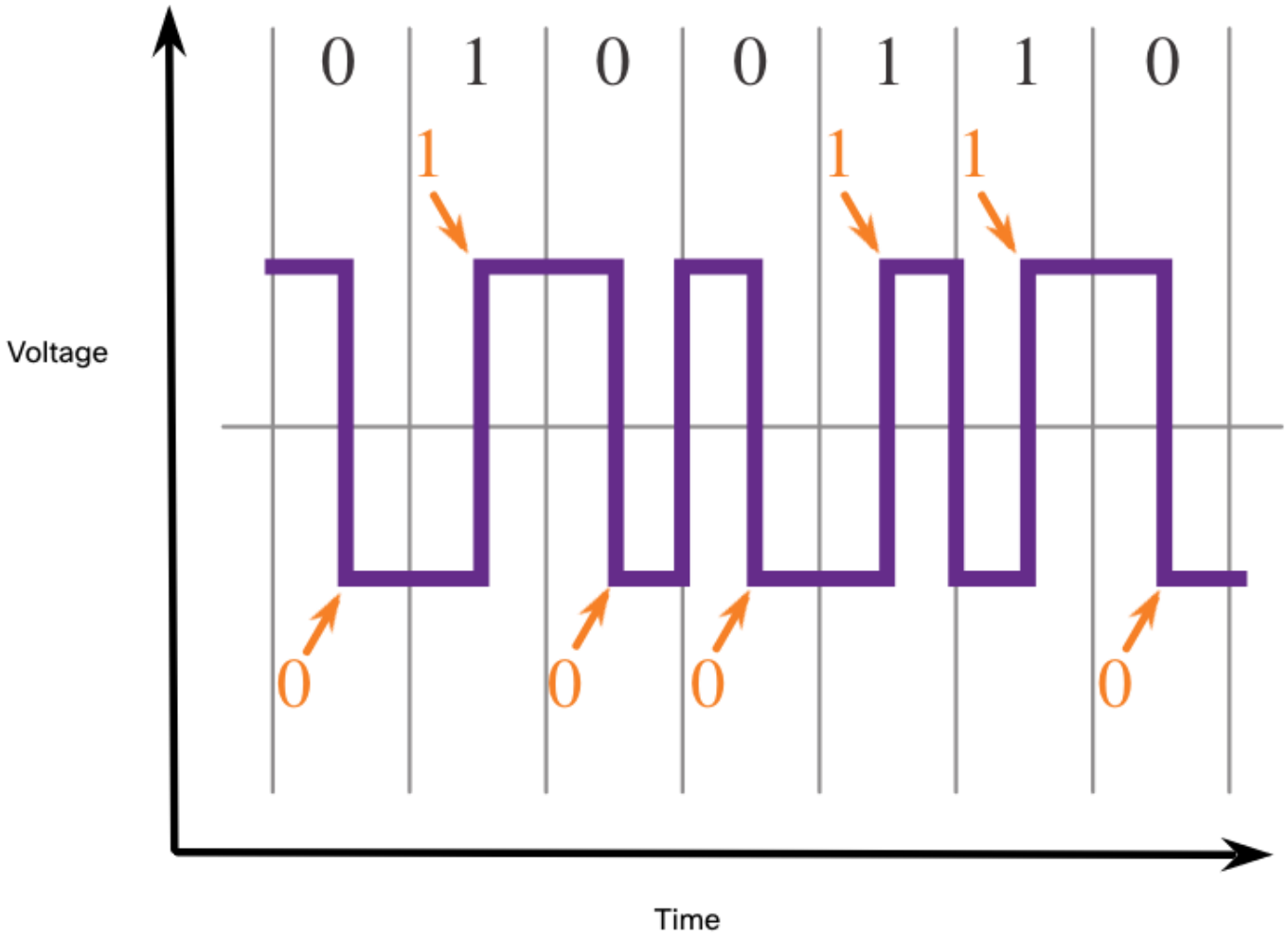
Digital signals: Encoding

- Consider the data stream 0 1 0 0 1 1 0 0 0 1 1
- This data can be encoded using a number of encoding schemes
  - Some have two voltage levels (binary)
  - Some have three voltage levels (ternary)
  - Some have positive and negative voltages (bipolar)
  - Some have only 0 and positive voltages (unipolar)



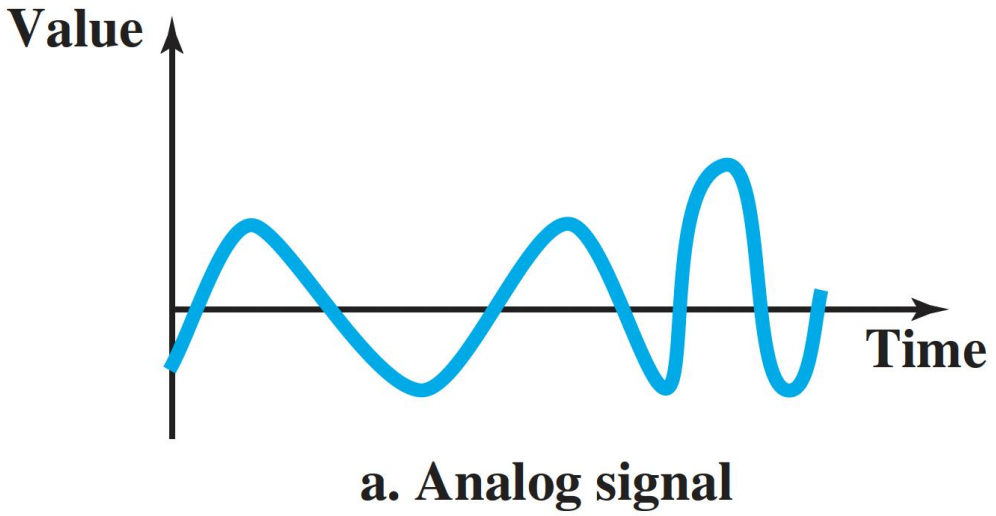
Manchester Encoding

- Used in Ethernet networks at low speeds (10 Mbps)
- 0 represented using a transition from high to low
- 1 represented using a transition from low to high
- Transition occurs at the middle of the bit period



Analog signals

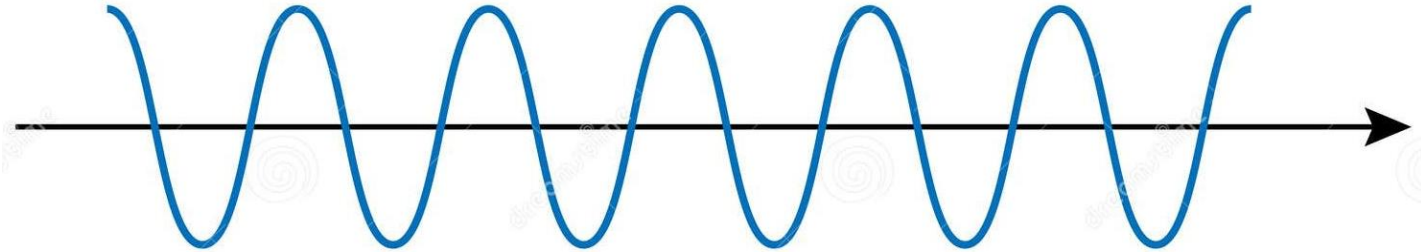
- An analog signal is something that is varying continuously in time and amplitude
- The amplitude refers to the strength of the signal
- The amplitude varies continuously (can take any value between an upper limit and a lower limit)



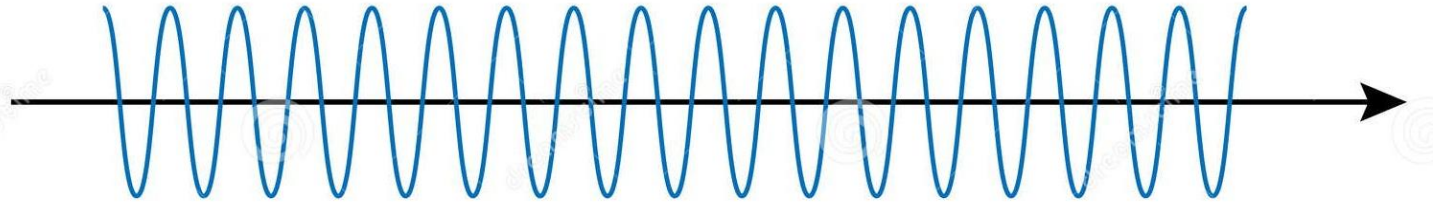
a. Analog signal

Frequency of an analog signal

- Frequency indicates how fast the signal is changing
- A Low frequency signal changes slowly with time



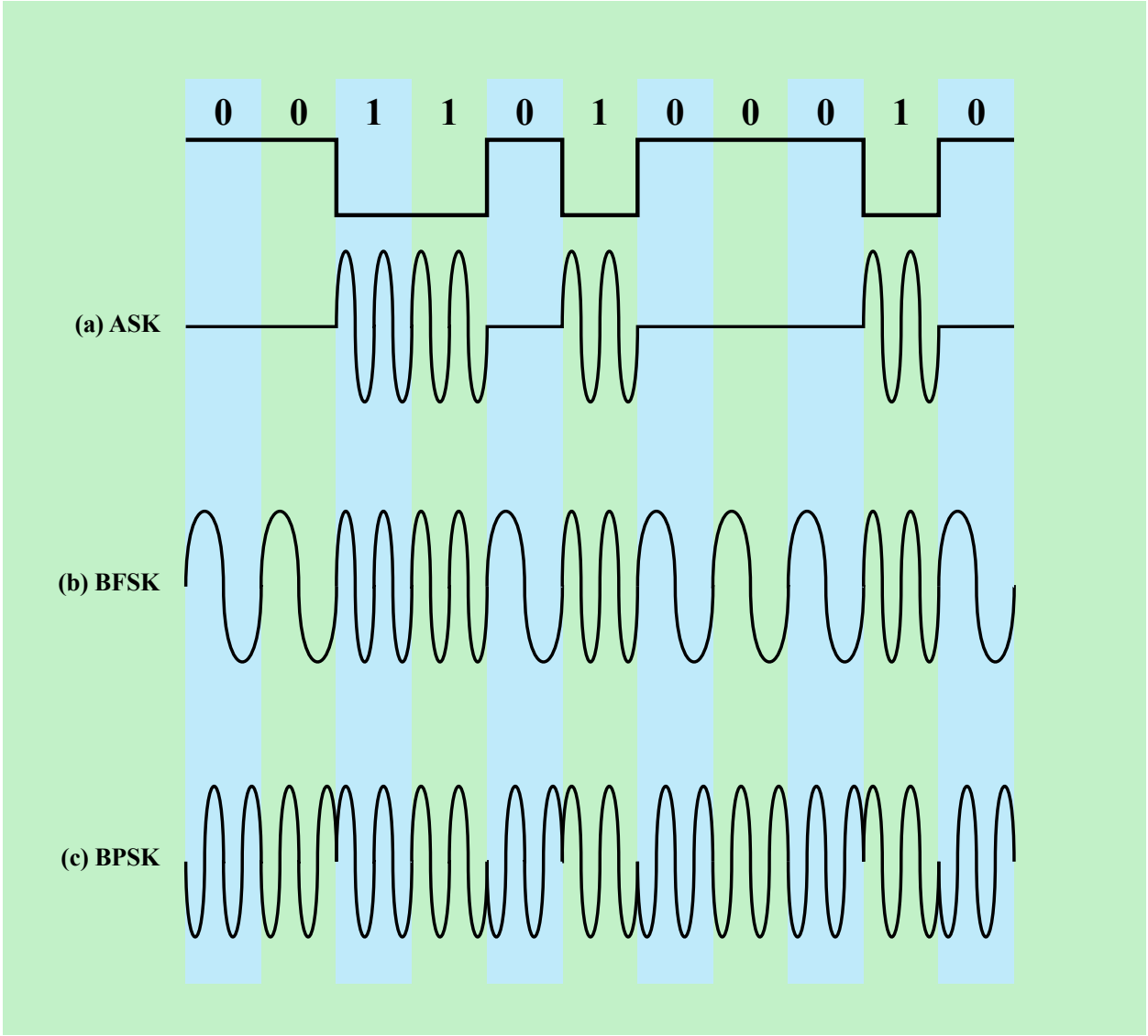
- A High frequency signal changes fast with time



- Frequency is expressed in cycles per second or Hertz (Hz)

Analog signals: Modulation

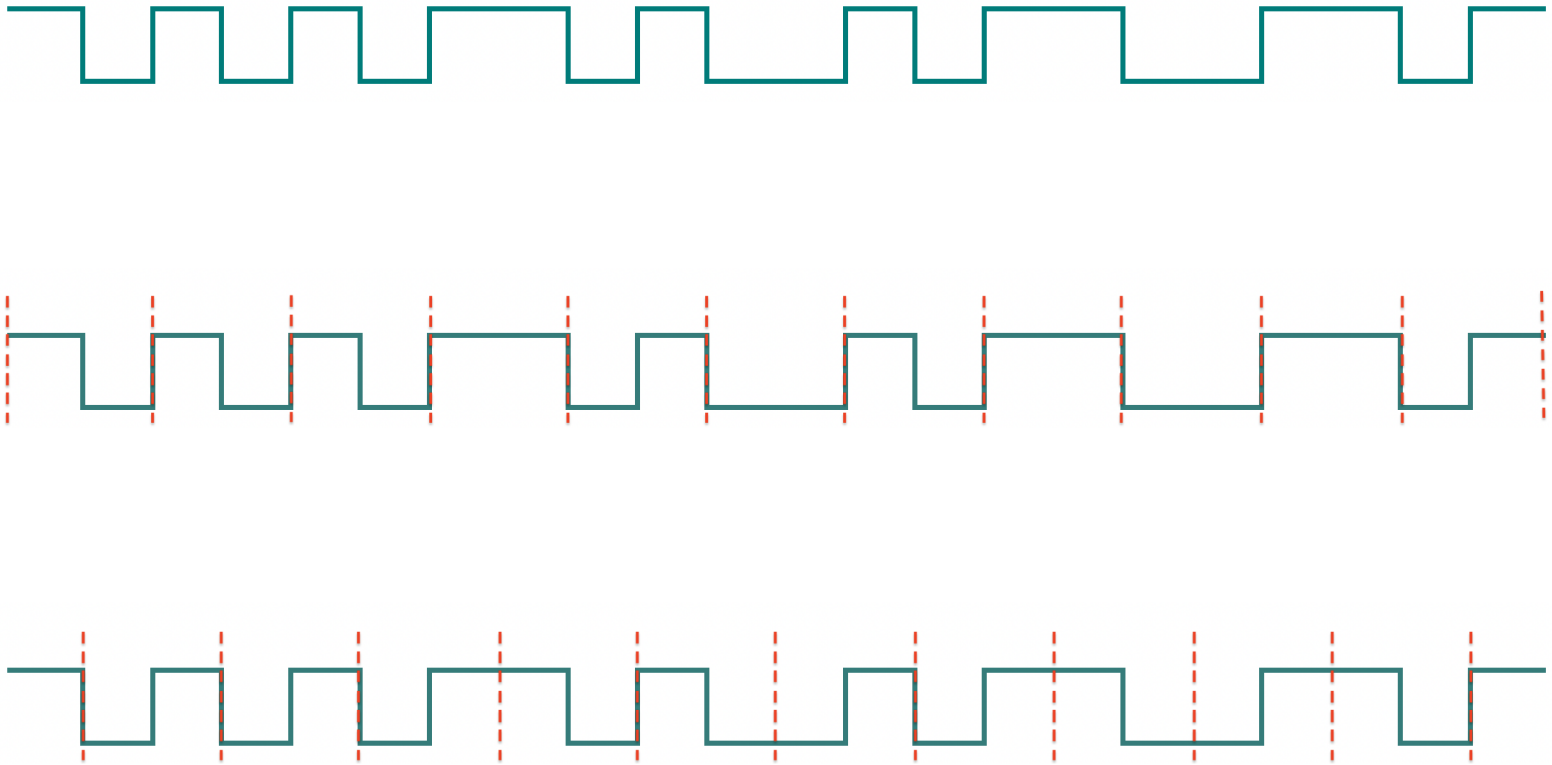
- Consider the data stream 0 0 1 1 0 1 0 0 0 1 0
- An analog signal is created using a modulation scheme
  - Amplitude modulation (ASK)
    - 1 and 0 represented using two different amplitude levels
  - Frequency modulation (BFSK)
    - 1 and 0 represented using two different frequencies
  - Phase modulation (BPSK)
    - 1 and 0 represented using two different phases (0 and +180)



Review questions

- What is the main function or responsibility of the physical layer?
- What is the relationship between data and signals?
- What are the two types of signals?
- What is encoding?
- What is modulation?
- The following waveform shows the signal on an Ethernet link. It has been encoded using Manchester encoding. Identify the bit periods and determine the data sequence.

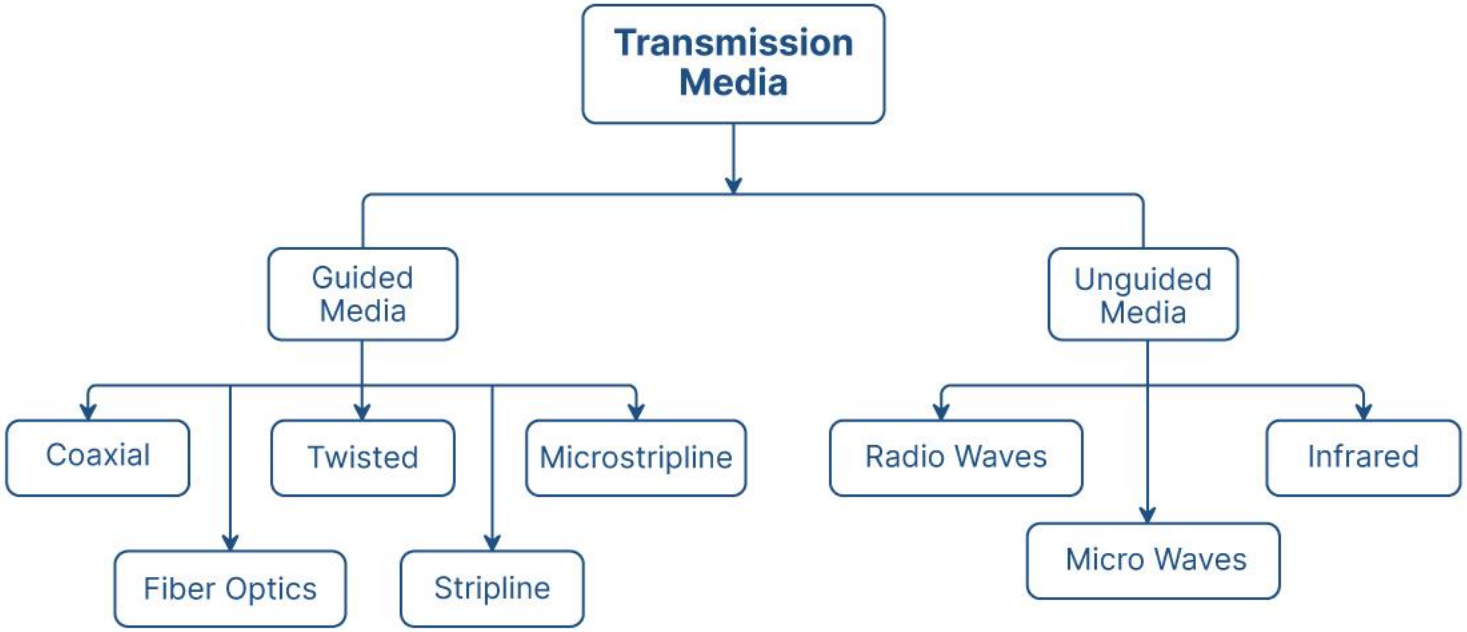




Which of these two figures show the correct bit periods?

Hint: There should be a transition at the middle of each bit period.

Transmission Media



Design Factors Determining Data Rate and Distance

- Bandwidth
  - Higher bandwidth gives higher data rate
- Transmission impairments
  - Impairments, such as attenuation, limit the distance
- Interference
  - Overlapping frequency bands can distort or wipe out a signal
- Number of receivers
  - More receivers introduces more attenuation (in guided media)

Transmission Characteristics of Guided Media

	Frequency Range	Typical Attenuation	Typical Delay	Repeater Spacing
Twisted pair (with loading)	0 to 3.5 kHz	0.2 dB/km @ 1 kHz	50 μs/km	2 km
Twisted pairs (multipair cables)	0 to 1 MHz	0.7 dB/km @ 1 kHz	5 μs/km	2 km
Coaxial cable	0 to 500 MHz	7 dB/km @ 10 MHz	4 μs/km	1 to 9 km
Optical fiber	186 to 370 THz	0.2 to 0.5 dB/km	5 μs/km	40 km

Characteristics of Copper Cabling

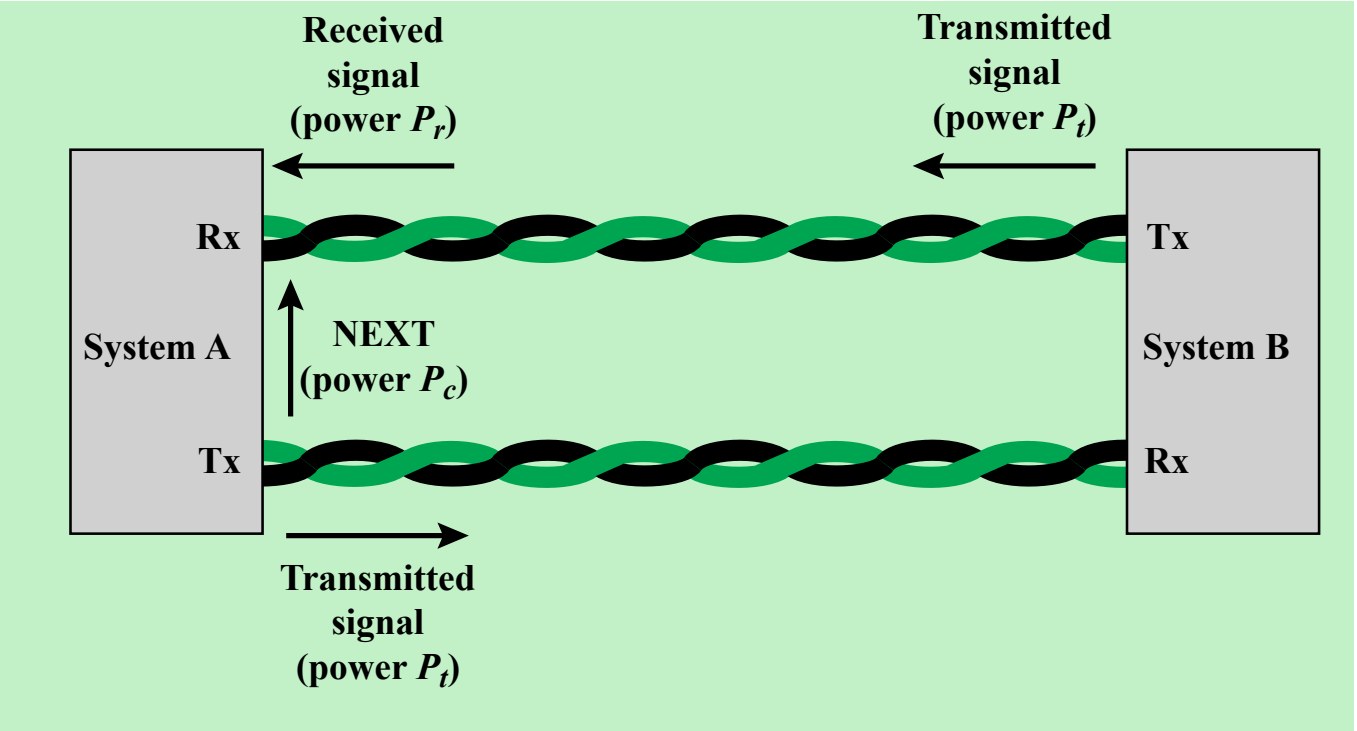
- Copper cabling is the most common type used in networks today
- It is inexpensive, easy to install, and copper has low resistance to electrical current flow
- Limitations: Attenuation, Interference and Crosstalk
- Attenuation – signal becomes weaker as it goes away from the source
  - Solution: Limit the cable length
- Interference – the electrical signal is susceptible to electromagnetic interference (EMI) from external signals
  - Solution: Using metallic shielding around the cables
- Crosstalk – coupling of signal from one pair of conductors to another
  - Solution: Twisting wire pairs, shield individual pairs

Attenuation / Near-End Crosstalk (NEXT)

- Attenuation
  - The signal gets weaker as it goes away from the source
  - For guided media (e.g., twisted-pair wire, optical fiber), attenuation is generally exponential and therefore expressed in decibels per unit distance
  - A received signal must have sufficient strength so that the electronic circuitry in the receiver can detect and interpret the signal
- NEXT (Near End Cross Talk)
  - A measure of coupling of signal from one pair of conductors to another
  - Near end: coupling at the end where the signal is transmitted

Attenuation / Near-End Crosstalk (NEXT)

- Attenuation and NEXT are usually expressed in dB (Decibel)
- Attenuation usually expressed in dB per unit distance (e.g. dB/m)
- $A_{dB} = 10 \log_{10} \frac{P_t}{P_r}$
- $NEXT_{dB} = 10 \log_{10} \frac{P_t}{P_c}$
- Media with lower attenuation are preferred
- Media with higher NEXT are preferred

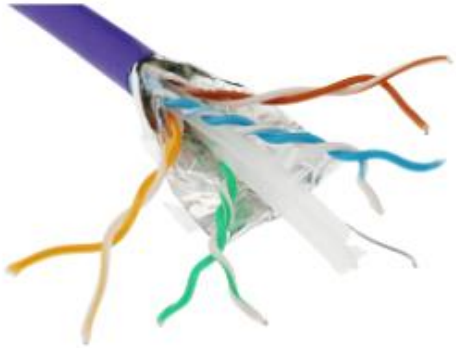


Types of Copper Cabling

- Unshielded Twisted Pair (UTP)
- Shielded Twisted Pair (STP)
- Coaxial cable



Unshielded Twisted-Pair (UTP) Cable



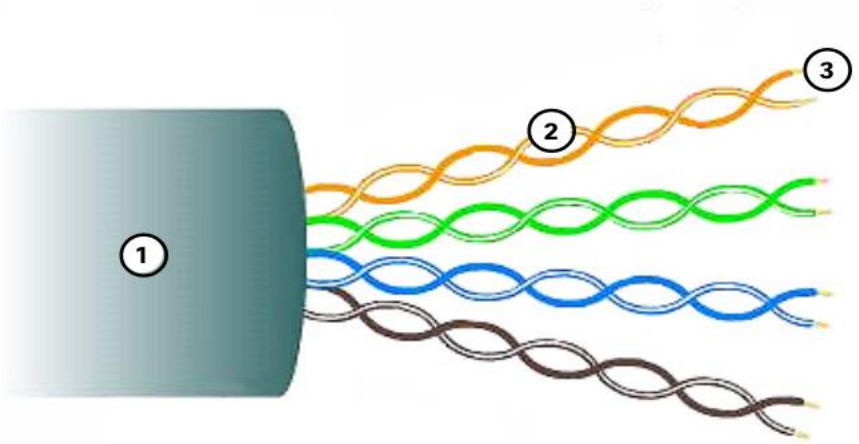
Shielded Twisted-Pair (STP) Cable



Coaxial Cable

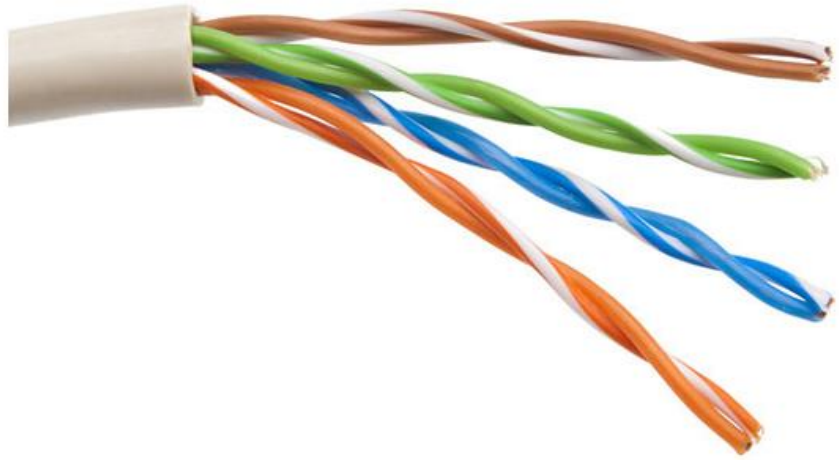
Unshielded twisted pair (UTP)

- UTP has four pairs of color-coded copper wires twisted together and encased in a flexible plastic sheath
- No shielding is used (Unshielded)
- Key Characteristics of UTP
  1. The outer jacket protects the copper wires from physical damage
  2. Twisting of the pairs help to reduce interference
  3. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair



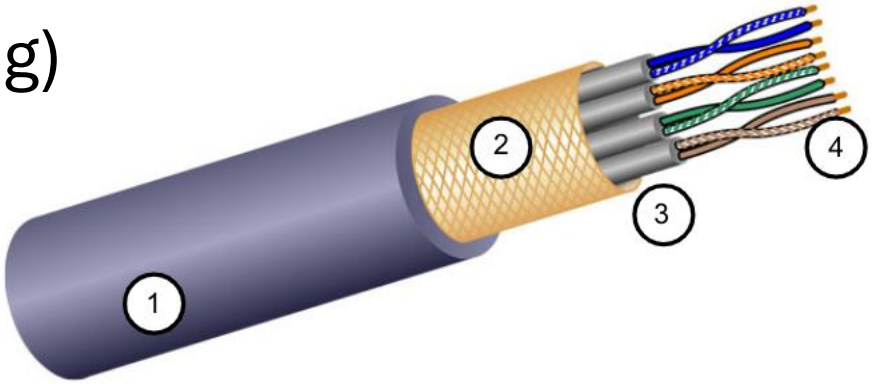
Limiting Crosstalk

- UTP relies on the following properties to limit crosstalk:
- Cancellation
  - When a pair of wires is twisted together, the magnetic fields effectively cancel each other and outside interference (EMI)
- Variation in twists per foot in each pair
  - Each pair is twisted a different amount



Shielded twisted pair (STP)

- Better noise protection than UTP
- More expensive than UTP
- Slightly harder to install than UTP (bending)
- Key Characteristics of STP
    1. Outer jacket protects the copper wires from physical damage
    2. Braided or foil shield provides protection from interference
    3. Foil shield for each pair of wires (if present) also reduces interference and crosstalk
    4. Color-coded plastic insulation electrically isolates the wires from each other and identifies each pair



UTP/STP Connectors



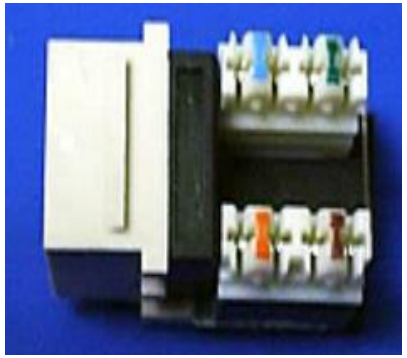


RJ-45 Connector



Poorly terminated UTP cable





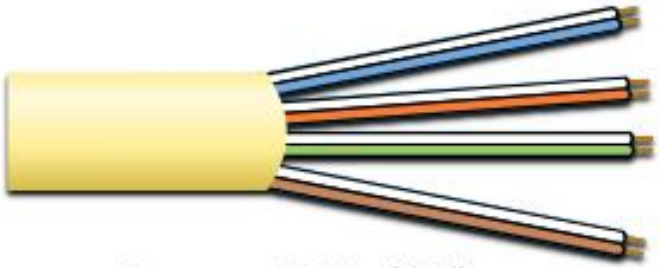
RJ-45 Socket



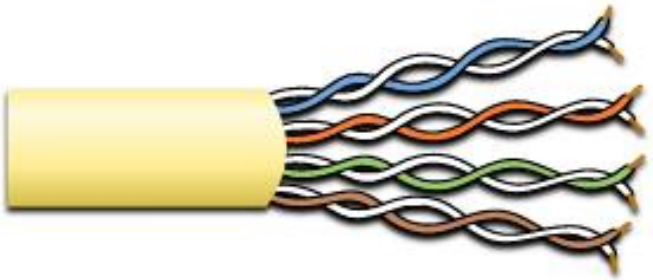
Properly terminated UTP cable

UTP cable standards

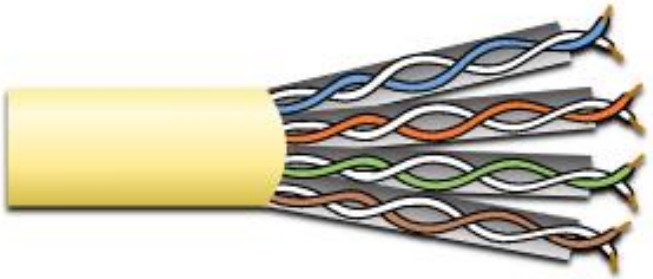
- Standards for UTP are established by the TIA/EIA. TIA/EIA-568 standardizes elements like:
  - Cable Types
  - Cable Lengths
  - Connectors
  - Cable Termination
  - Testing Methods
- Electrical standards for copper cabling are established by the IEEE, which rates cable according to its performance. Examples include:
  - Category 3
  - Category 5 and 5e
  - Category 6



Category 3 Cable (UTP)



Category 5 and 5e Cable (UTP)



Category 6 Cable (UTP)

Twisted pair categories and classes

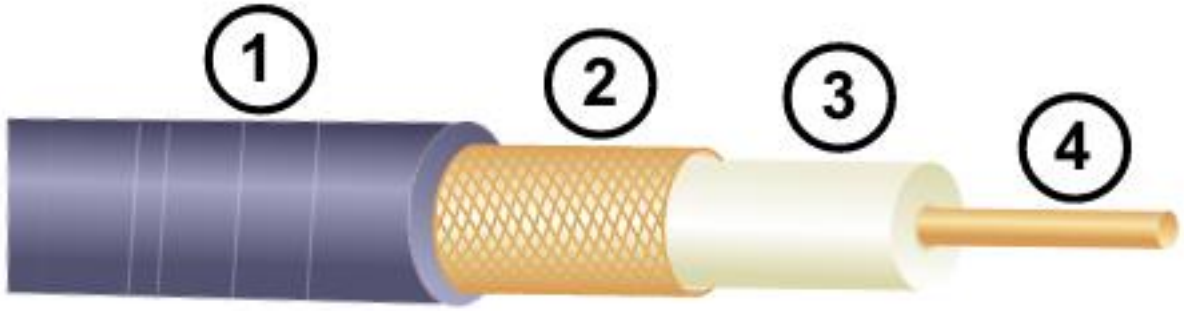
UTP = Unshielded twisted pair
FTP = Foil twisted pair
S/FTP = Shielded/foil twisted pair

ACR = Attenuation to crosstalk ratio

	Category 5e Class D	Category 6 Class E	Category 6A Class E_A	Category 7 Class F	Category 7_A Class F_A
Bandwidth	100 MHz	250 MHz	500 MHz	600 MHz	1,000 MHz
Cable Type	UTP	UTP/FTP	UTP/FTP	S/FTP	S/FTP
Insertion loss (dB)	24	21.3	20.9	20.8	20.3
NEXT loss (dB)	30.1	39.9	39.9	62.9	65
ACR (dB)	6.1	18.6	19	42.1	44.1

Coaxial cable

- Consists of the following
  1. Outer cable jacket to prevent minor physical damage
  2. A woven copper braid, or metallic foil, acts as the second wire in the circuit and as a shield for the inner conductor.
  3. A layer of flexible plastic insulation
  4. A copper conductor is used to transmit the electronic signals.
- There are different types of connectors used with coax cable.
- Commonly used in wireless installations - attach antennas to wireless devices



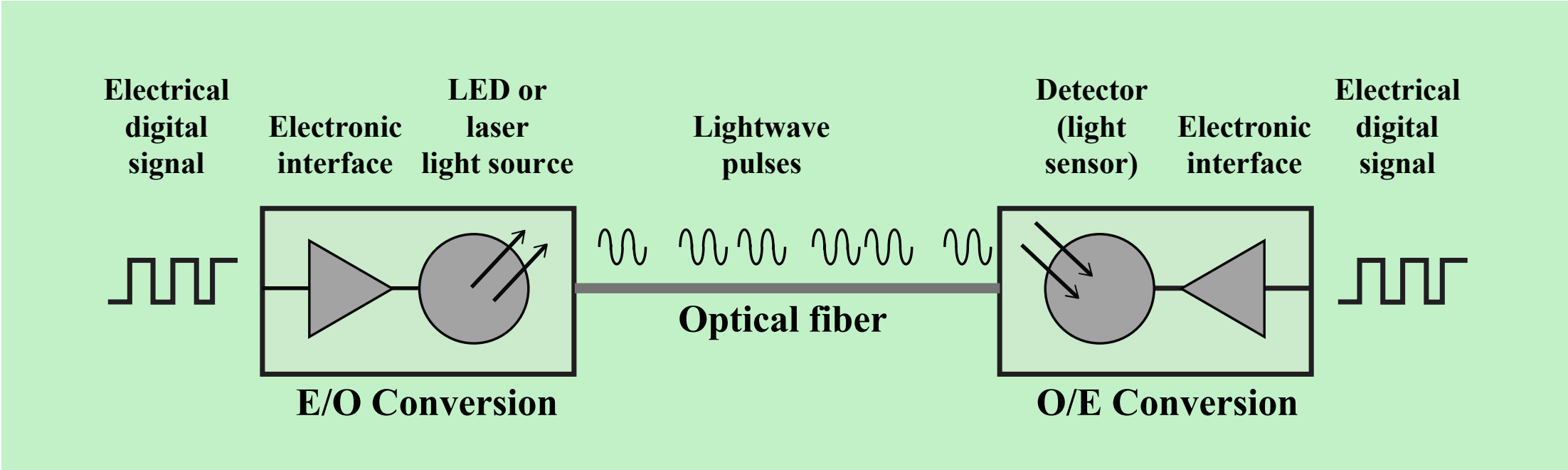


Review questions

- What are the three main limitations in UTP cable?
- What are the differences between UTP and STP cables?
- What is attenuation?
- What is electro-magnetic interference (EMI)?
- What is crosstalk?
- What is the difference between a crossover cable and a straight-through cable?
- Why are crossover cables considered obsolete now?

Optical communication

- Optical communication using optical fiber requires
  - Electrical to optical conversion at the transmitter end
  - Optical to electrical conversion at the receiver end

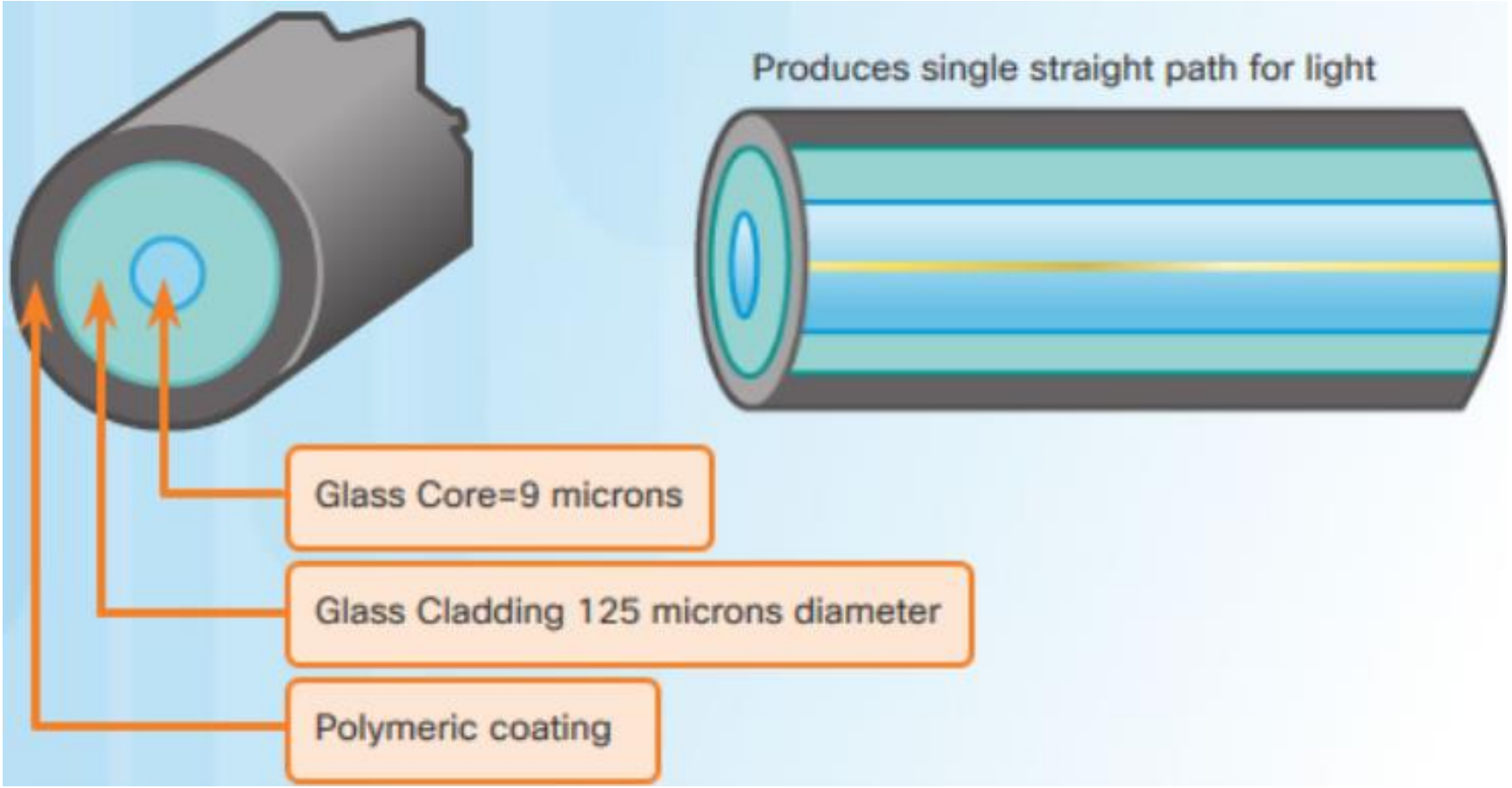


Properties of Fiber Optic Cabling

- Transmits data over longer distances at higher bandwidth than any other networking media
- Less susceptible to attenuation, and completely immune to EMI/RFI
- Made of flexible, extremely thin strands of very pure glass
- Uses a laser or LED to encode bits as pulses of light
  - 1 – presence of light
  - 0 – absence of light
- The fiber-optic cable acts as a wave guide to transmit light between the two ends with minimal signal loss

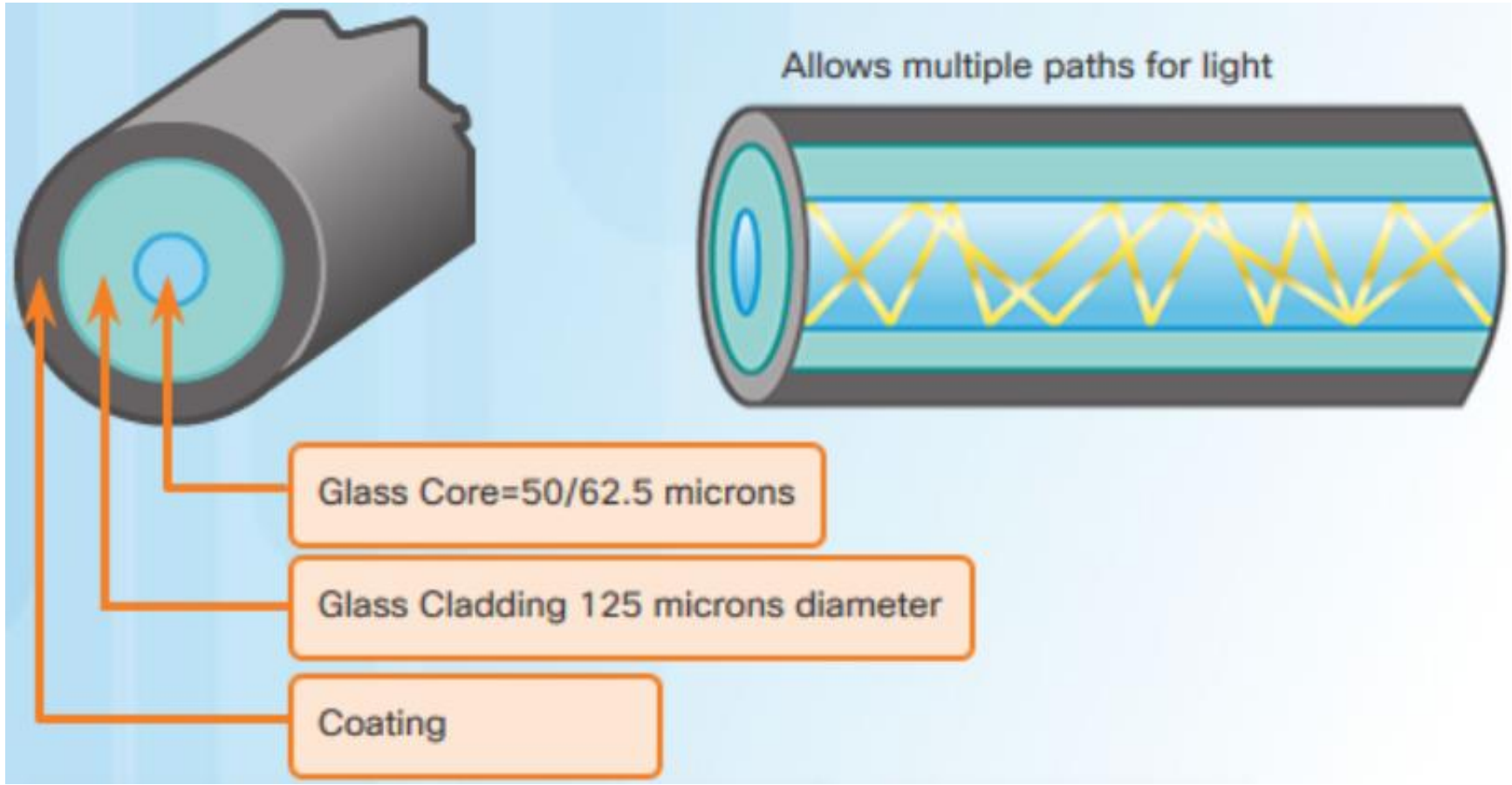
Types of Fiber – Single-Mode (SMF)

- Very small core
- Uses expensive lasers
- Long-distance applications
- Typically 5-10km
- 40km/80km possible



Types of Fiber – Multimode (MMF)

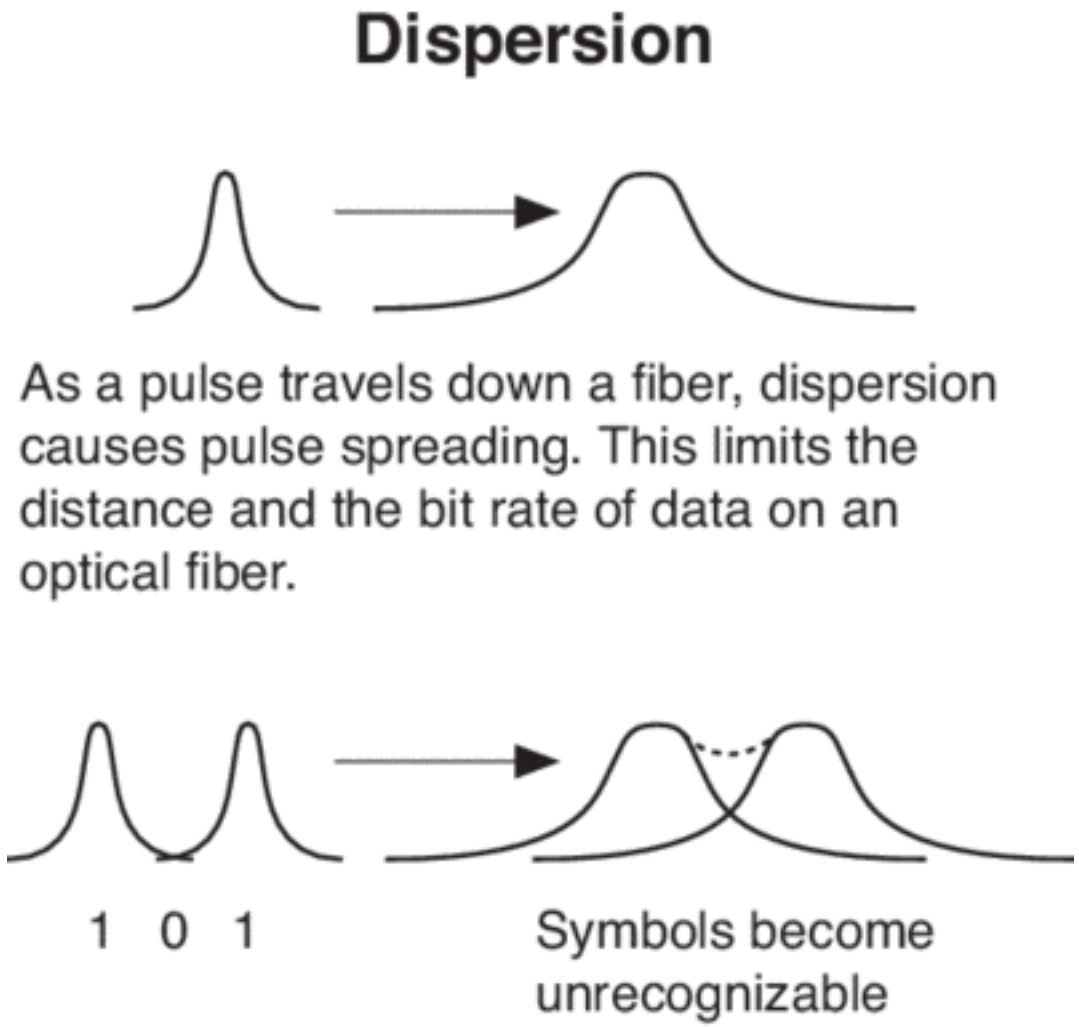
- Larger core
- Uses less expensive LEDs
- LEDs emit light at different angles
- maximum distance for MMF is 550 meters



- Due to higher dispersion (spreading of a light pulse) than SMF

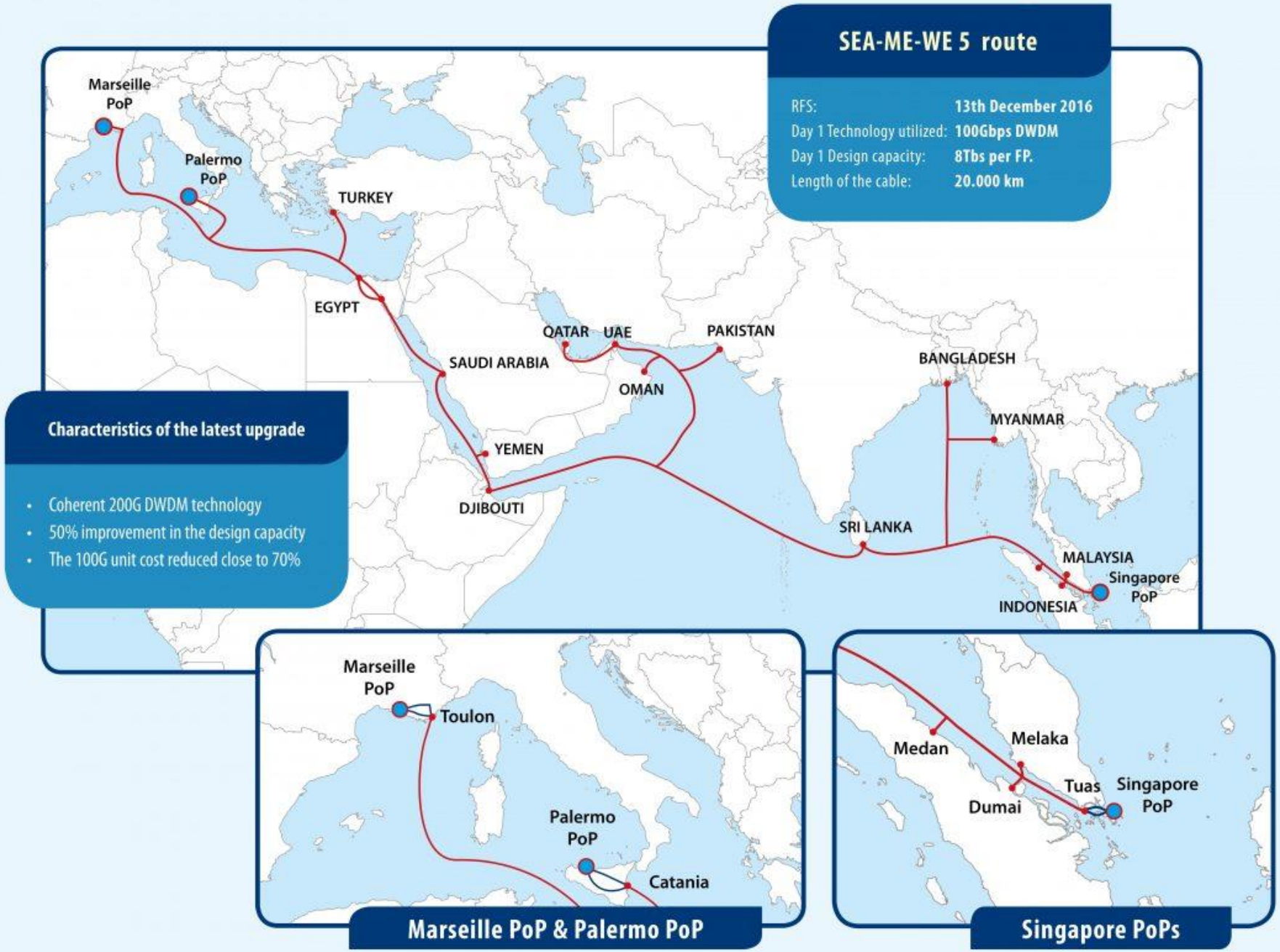
Dispersion

- Dispersion refers to the spreading out of a light pulse over time.
- Increased dispersion means increased loss of signal strength.
- MMF has greater dispersion than SMF.
- The maximum cable distance for MMF is 550 meters.

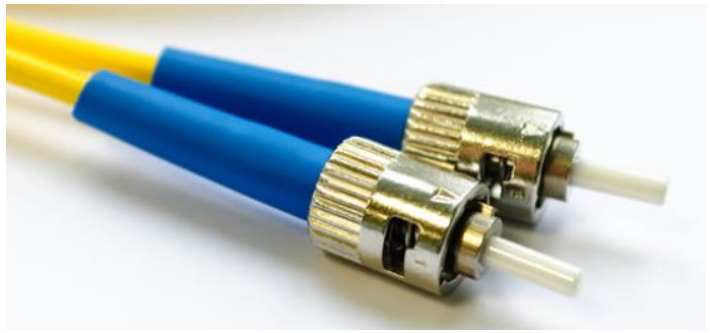


Fiber Optic Cabling usage

- Enterprise Networks - Used for backbone cabling applications and interconnecting infrastructure devices
- Fiber-to-the-Home (FTTH) - Used to provide always-on broadband services to homes and small businesses
- Long-Haul Networks - Used by service providers to connect countries and cities
- Submarine Cable Networks - Used to provide reliable high-speed, high-capacity solutions capable of surviving in harsh undersea environments at up to transoceanic distances


SEA-ME-WE 5 route
RFS: 13th December 2016
Day 1 Technology utilized: 100Gbps DWDM
Day 1 Design capacity: 8Tbs per FP.
Length of the cable: 20.000 km
Marseille PoP
Palermo PoP
TURKEY
EGYPT
QATAR
UAE
PAKISTAN
SAUDI ARABIA
OMAN
BANGLADESH
MYANMAR
YEMEN
DJIBOUTI
SRI LANKA
MALAYSIA
Singapore PoP
INDONESIA
Characteristics of the latest upgrade
• Coherent 200G DWDM technology
• 50% improvement in the design capacity
• The 100G unit cost reduced close to 70%
Marseille PoP
Toulon
Palermo PoP
Catania
Marseille PoP & Palermo PoP
Medan
Melaka
Tuas
Singapore PoP
Dumai
Singapore PoPs

Fiber-Optic Connectors



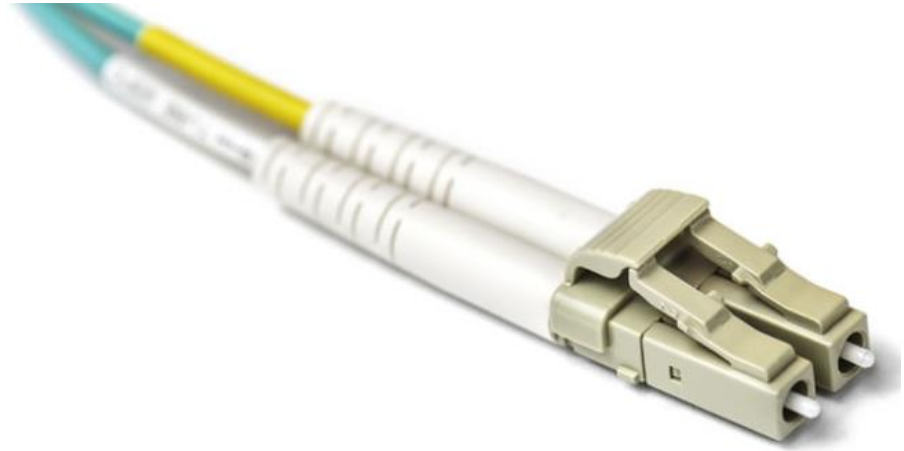
Straight-Tip (ST) Connectors



Lucent Connector (LC) Simplex Connectors



Subscriber Connector (SC) Connectors



Duplex Multimode LC Connectors

Fiber Patch Cords



SC-SC MM Patch Cord



LC-LC SM Patch Cord



ST-LC MM Patch Cord



ST-SC SM Patch Cord

A yellow jacket is for single-mode fiber cables and orange (or aqua) for multimode fiber cables.

Optical Fiber - Benefits

- Greater capacity
  - Data rates of hundreds of Gbps over tens of kilometers have been demonstrated
- Smaller size and lighter weight
  - Considerably thinner than coaxial or twisted pair cable
  - Reduces structural support requirements
- Lower attenuation
- Electromagnetic isolation
  - Not vulnerable to interference, impulse noise, or crosstalk
  - High degree of security from eavesdropping
- Greater repeater spacing
  - Lower cost and fewer sources of error



Frequency Utilization for Fiber Applications

Wavelength (in vacuum) range (nm)	Frequency Range (THz)	Band Label	Fiber Type	Application
820 to 900	366 to 333		Multimode	LAN
1280 to 1350	234 to 222	S	Single mode	Various
1528 to 1561	196 to 192	C	Single mode	WDM
1561 to 1620	192 to 185	L	Single mode	WDM

WDM = wavelength division multiplexing

Fiber versus Copper

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 100 Gbps	10 Mb/s - 800 Gbps
Distance	Relatively short (1 – 100m)	Relatively long (1m – 100km)
Attenuation (per 100m)	High (22dB @ 100MHz)	Low (0.03 to 0.4 dB)
Immunity to EMI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Low	High
Installation skills required	Low	High
Safety precautions	Low	High

Review questions

- What are the differences between singlemode fiber and multimode fiber?
- What are the differences between the characteristics such as (Attenuation, EMI and Crosstalk) of UTP cabling and Fiber-Optic cabling?
- What are the benefits of Fiber-Optic cabling?
- What is the range of wavelengths of the light signals used in the L band for single mode fiber?

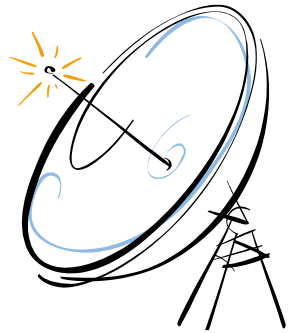
Properties of Wireless Media

- Free space carries electromagnetic signals representing 1s and 0s
- Wireless media becoming more popular over wired for LANs
  - Provides the greatest mobility option
- Some limitations of wireless:
  - Coverage area - Signal becomes weaker much faster than in wired media
  - Interference - Wireless signal can be disrupted by many common devices
  - Security - within the signal coverage area, anyone can connect unless protected
  - Shared medium - WLANs operate in half-duplex which means
    - only one device can transmit at any given time
    - Many users accessing the WLAN simultaneously results in reduced bandwidth for each user

Wireless Transmission Frequencies

- Wireless LANs (Wi-Fi)
  - 2.4 GHz band: Older Wi-Fi standards
  - 5 GHz band: Used by newer Wi-Fi standards such as 802.11a, 802.11n, 802.11ac, and 802.11ax.
  - 6 GHz band: Used by newer Wi-Fi standars (Wi-Fi 6 and later)
- Bluetooth
  - 2.4 GHz band: Specifically, Bluetooth operates in the range of 2.402 GHz to 2.480 GHz.
  - This is a globally unlicensed Industrial, Scientific, and Medical (ISM) band
- IrDA (Infrared Data Association)
  - Unlike Wi-Fi and Bluetooth, IrDA uses light waves
  - Typically around 875 nm (nanometers) to 900 nm wavelength

Antennas

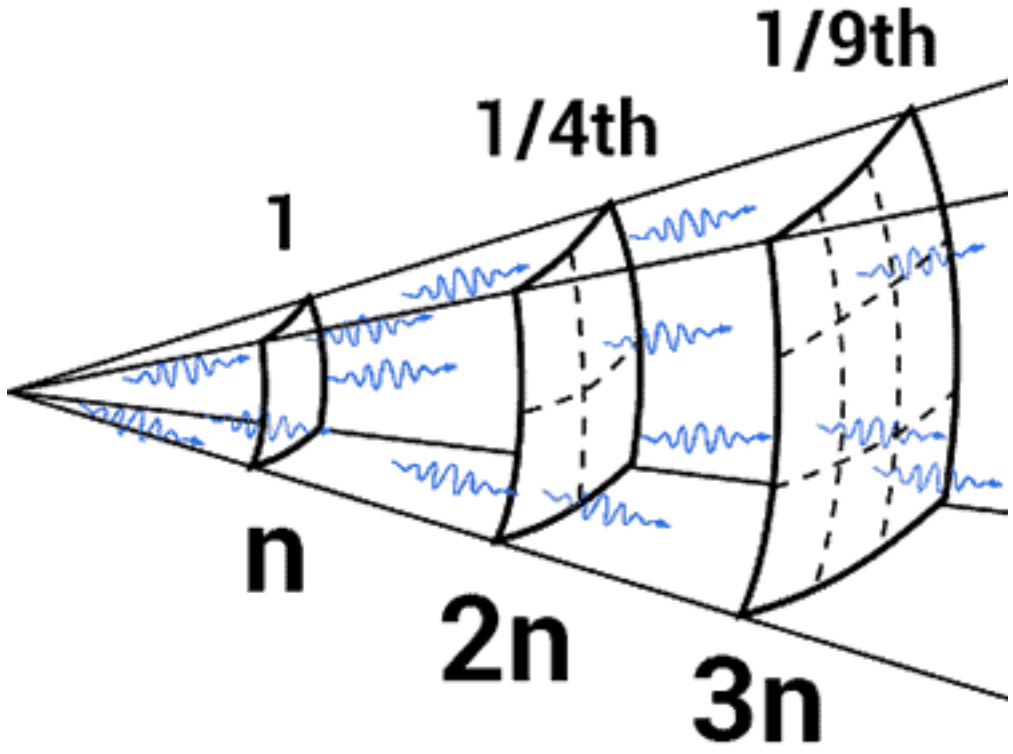


- Electrical conductor or a system of conductors used to radiate or collect electromagnetic energy
- Electrical energy from the transmitter is converted into electromagnetic energy by the antenna and radiated into the surrounding environment
- Reception occurs when the electromagnetic signal intersects the receiver antenna
- In two-way communication, the same antenna can be used for both transmission and reception



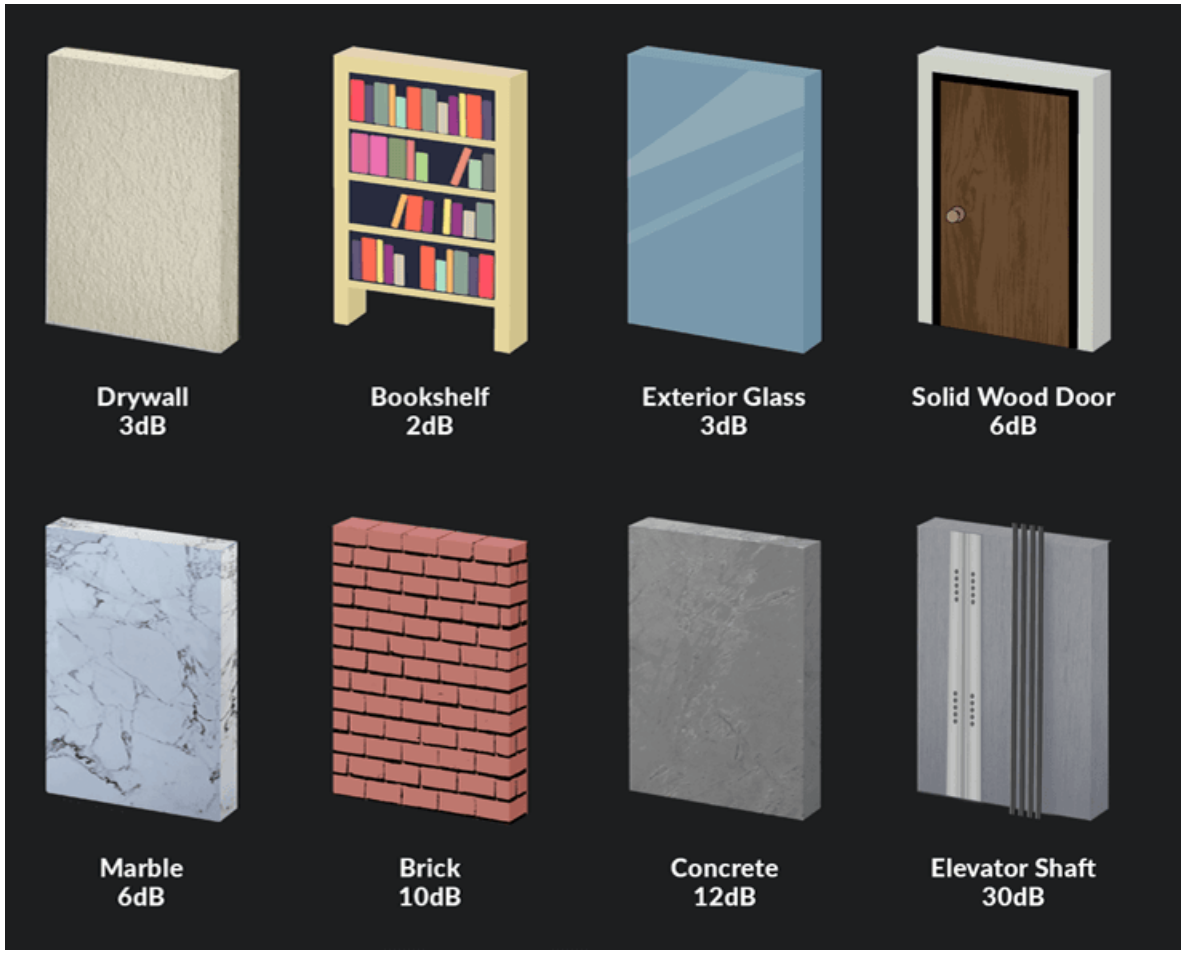
Wireless signals - attenuation

- Wireless signals get attenuated faster than signals on guided media (copper, fiber)
  - Signal strength inversely proportional to the square of distance from the source
  - Signal strength at 4m distance is 1/16 of signal strength at 1m
  - Signal strength at 10m distance is 1/100 of signal strength at 1m
- This is called free space loss or inverse square law



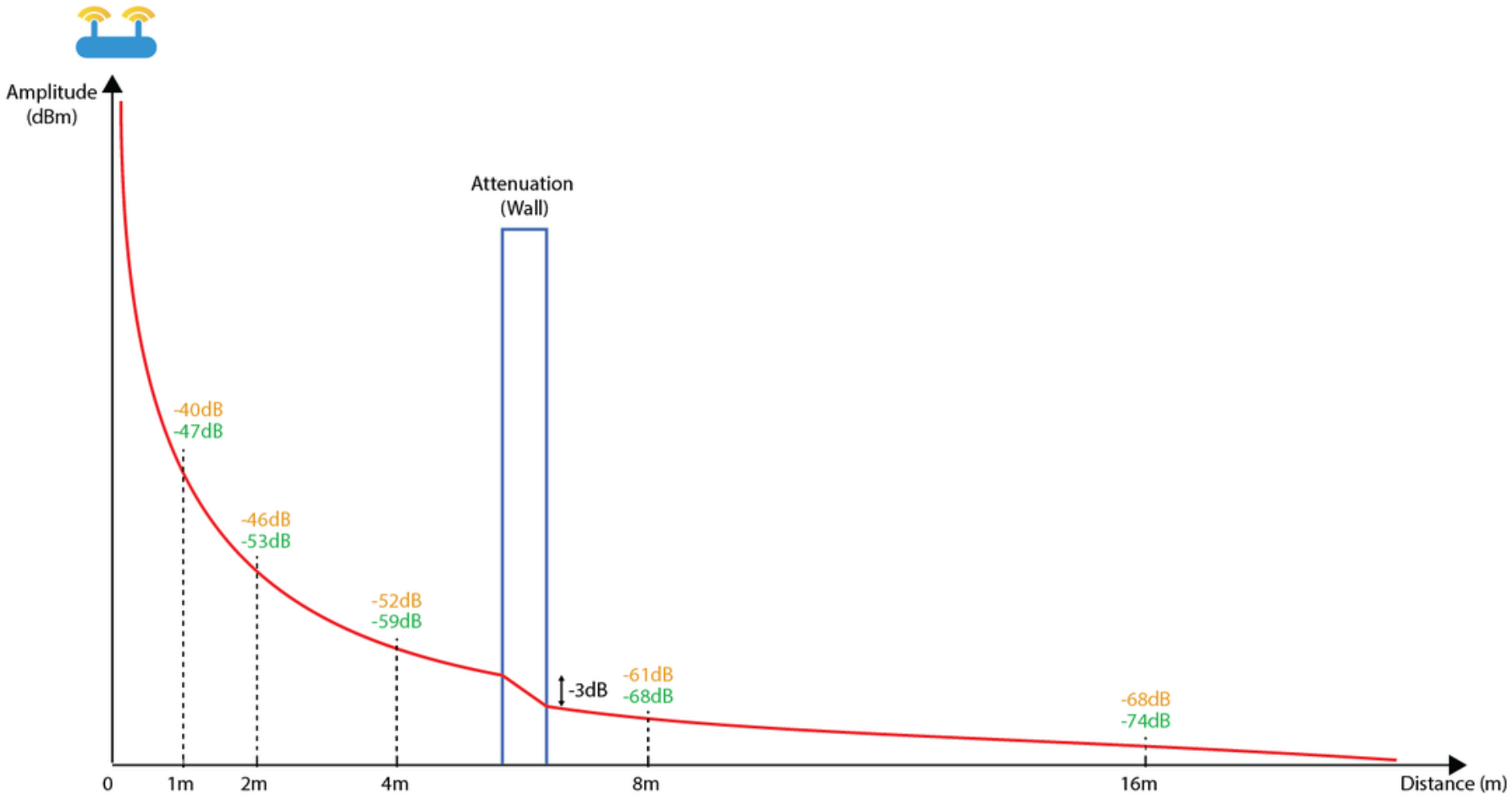
Wireless signals – attenuation (2)

- Wireless signals get attenuated due to obstacles between the transmitter and receiver
- Depends on the material and thickness of the obstacle in signal path
- Typical loss
  - Bookshelf 2dB (= 2/3)
  - Glass 3dB (= ½)
  - Solid wood door 6dB (= 1/4)
  - Brick wall 10dB (= 1/10)
  - Concrete 12dB (= 1/16)
  - Elevator shaft 30dB (=1/1000)



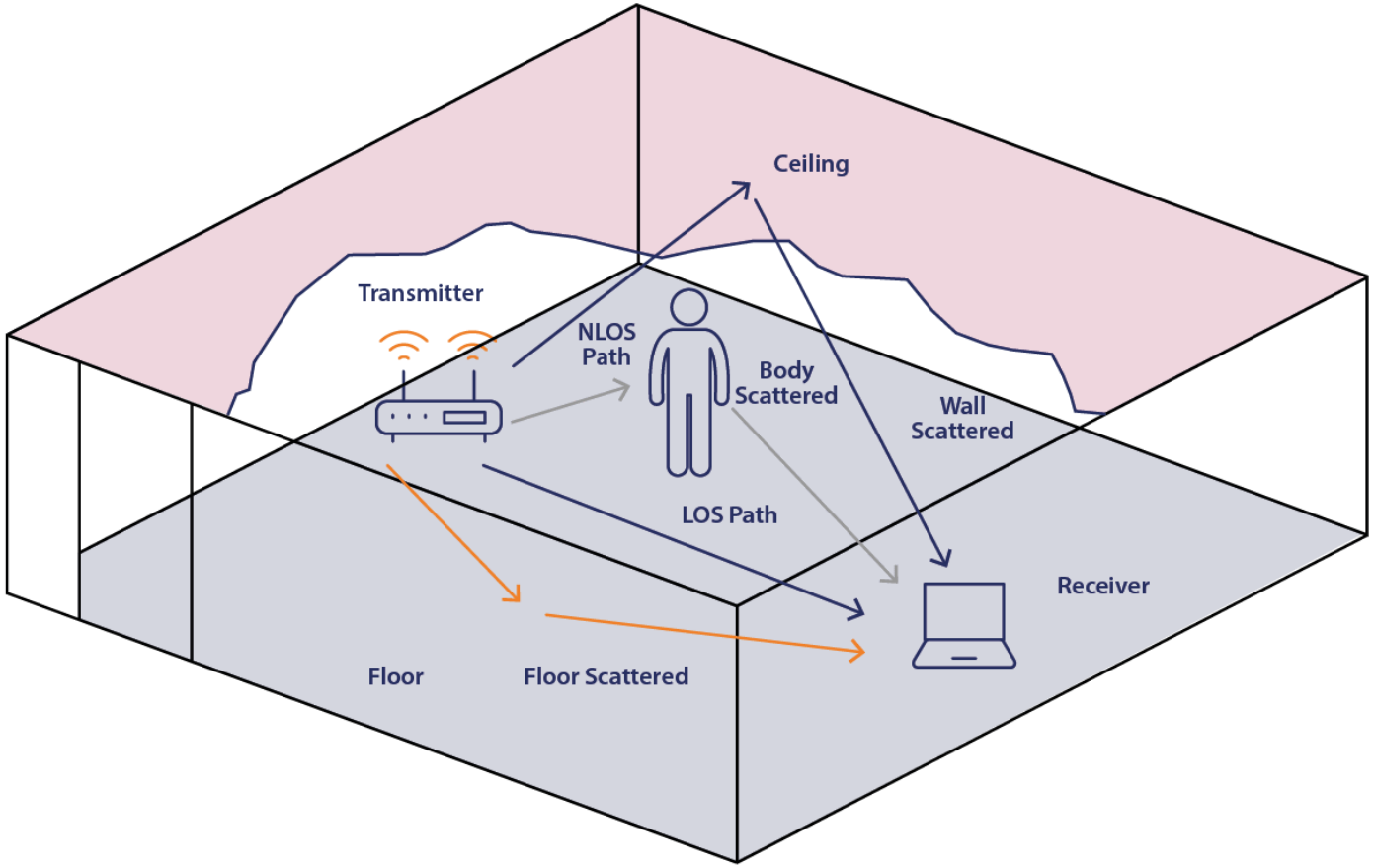
Source: ekahau.com

Wireless signals – attenuation (3)



Multipath interference

- Multiple interfering signals from reflections
- Each path has a different propagation delay from the transmitter



Infrared

- Achieved using transceivers that modulate noncoherent infrared light
- Transceivers must be within line of sight of each other directly or via reflection
- Does not penetrate walls
- No licensing is required
- No frequency allocation issues



Key terms and concepts

- Analog and digital signals
- Encoding
- Modulation
- Attenuation
- Interference
- Twisted Pair cable
- Single-mode and Multi-mode fiber
- Wireless media
- Transmission frequencies

References

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